

Problem 25.3

(a) It is equivalent to ionization of a hydrogen-like atom with $Z = 2$,

$$\mu = \frac{4m_p m}{4m_p + m} = \frac{m}{1.00003}$$

$$E = E_1(\text{H}) \frac{Z^2}{\eta} = -13.6\text{eV} \times \frac{4}{1.00003} = -54.4\text{eV}$$

(b) $79\text{eV} - 54.4\text{eV} = 24.6\text{eV}$

Problem 25.5 $\left(\rho = \frac{2Z}{na_0} r \right)$

$$\frac{\partial(R_{2,0}^2 r^2)}{\partial r} = 0 \Rightarrow \frac{d}{d\rho} [(2-\rho)^2 \rho^2 e^{-\rho}]$$

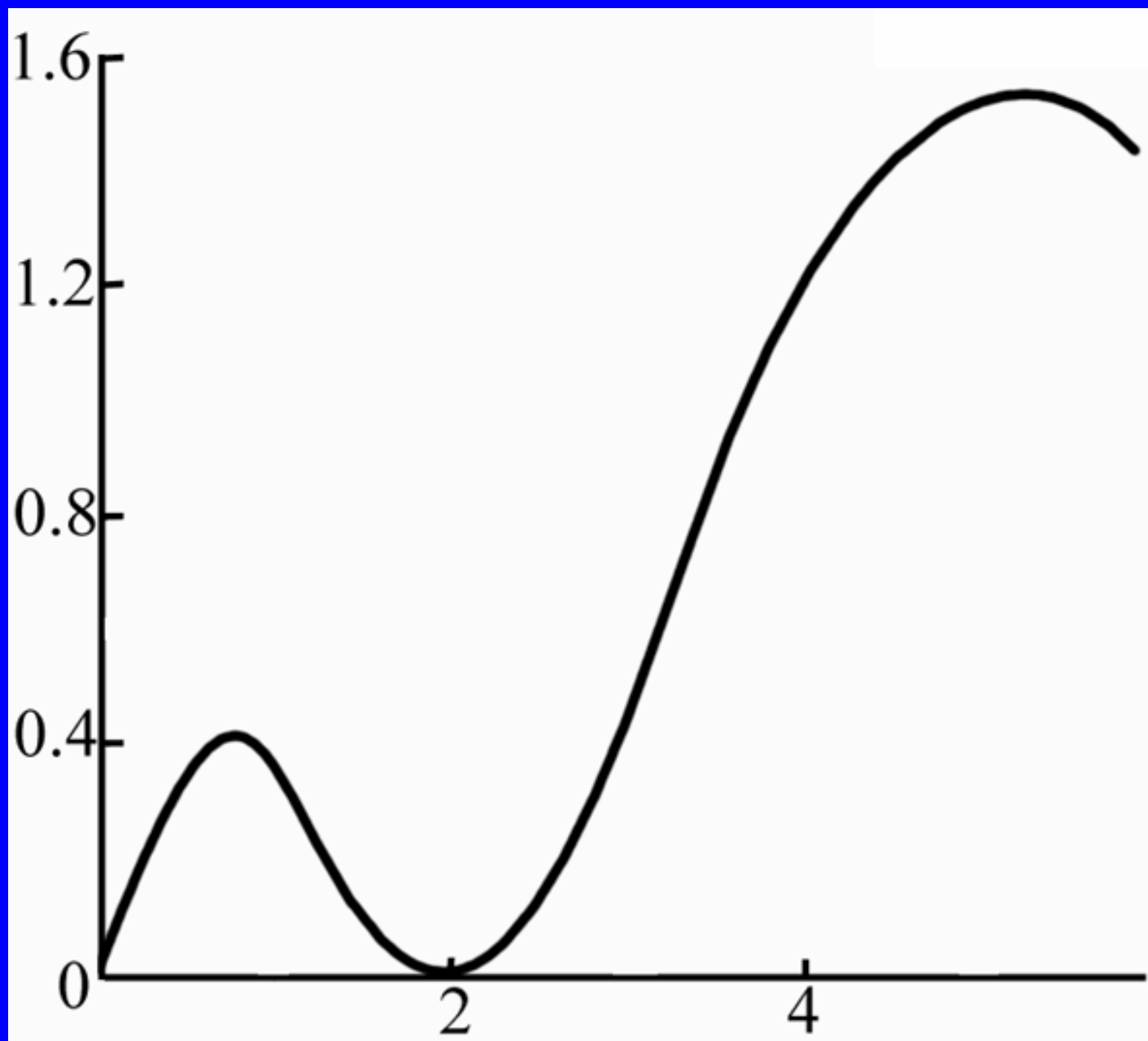
$$0 = -2(2-\rho)\rho^2 + 2\rho(2-\rho)^2 - (2-\rho)^2 \rho^2$$

$$\rho = 0, 2$$

$$-2\rho + 2(2-\rho) - (2-\rho)\rho = 0$$

$$\rho = 3 \pm \sqrt{5}$$

Finding the second derivative or plotting, we know $r = \left(3 \pm \sqrt{5}\right) \frac{a_0}{Z}$ are maxima.



Problem 25.7*

(a) Possible values are:

$$E_2, E_3; \quad 1 \cdot (1+1)\hbar^2 = 2\hbar^2; \quad 0, -\hbar$$

$$\frac{1}{2}R_{21}(r)Y_{10}(\theta, \varphi) - \frac{\sqrt{3}}{2}R_{31}(r)Y_{1,-1}(\theta, \varphi)$$

(b)

$$\begin{aligned}\langle E \rangle &= \int \left[\frac{1}{2} R_{21}(r) Y_{10}(\theta, \varphi) - \frac{\sqrt{3}}{2} R_{31}(r) Y_{1,-1}^*(\theta, \varphi) \right] \\ &\quad \hat{H} \left[\frac{1}{2} R_{21}(r) Y_{10}(\theta, \varphi) - \frac{\sqrt{3}}{2} R_{31}(r) Y_{1,-1}(\theta, \varphi) \right] d\tau \\ &= \int \left[\frac{1}{2} R_{21}(r) Y_{10}(\theta, \varphi) - \frac{\sqrt{3}}{2} R_{31}(r) Y_{1,-1}^*(\theta, \varphi) \right] \\ &\quad \left[\frac{1}{2} E_2 R_{21}(r) Y_{10}(\theta, \varphi) - \frac{\sqrt{3}}{2} E_3 R_{31}(r) Y_{1,-1}(\theta, \varphi) \right] d\tau \\ &= \frac{1}{4} E_2 + \frac{3}{4} E_3\end{aligned}$$

$$\langle \hat{L}^2 \rangle = 2\hbar^2$$

$$\begin{aligned}
\langle \hat{L}_z \rangle &= \int \left[\frac{1}{2} R_{21}(r) Y_{10}(\theta, \varphi) - \frac{\sqrt{3}}{2} R_{31}(r) Y_{1,-1}^*(\theta, \varphi) \right] \\
&\quad \boxed{\hat{L}_z} \left[\frac{1}{2} R_{21}(r) Y_{10}(\theta, \varphi) - \frac{\sqrt{3}}{2} R_{31}(r) Y_{1,-1}(\theta, \varphi) \right] d\tau \\
&= \int \left[\frac{1}{2} R_{21}(r) Y_{10}(\theta, \varphi) - \frac{\sqrt{3}}{2} R_{31}(r) Y_{1,-1}^*(\theta, \varphi) \right] \\
&\quad \left[-\frac{\sqrt{3}}{2} \boxed{(-\hbar)} R_{31}(r) Y_{1,-1}(\theta, \varphi) \right] d\tau \\
&= -\frac{3}{4} \hbar
\end{aligned}$$

Problem 25.10

(The problem has been corrected as “shortest wavelength”)

$$\lambda_{\min} = \frac{hc}{E_k} = \frac{12\,40\,\text{eV} \cdot \text{nm}}{35\,\text{keV}} = 35.4\,\text{pm}$$

Problem 25.11 For X-ray energy levels:

$$E_n = -\frac{(Z-1)^2 e_s^2}{2a_0} \frac{1}{n^2} = -E_1 \frac{1}{n^2}$$

For K_α , K_β , the frequencies are respectively

$$h\nu_\alpha = E_1 \left(\frac{1}{1^2} - \frac{1}{2^2} \right) = \frac{3}{4} E_1$$

$$h\nu_\beta = E_1 \left(\frac{1}{1^2} - \frac{1}{3^2} \right) = \frac{8}{9} E_1$$

So

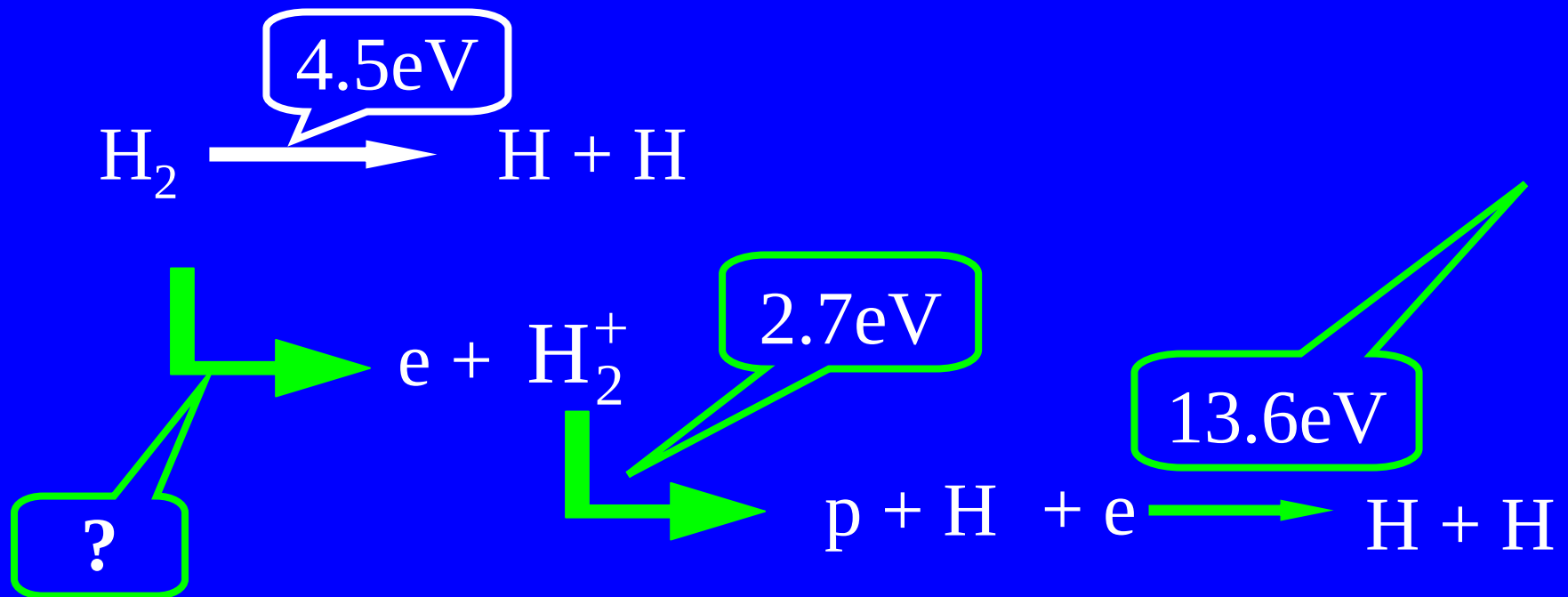
$$\nu_\beta = \frac{8}{9} \times \frac{4}{3} \nu_\alpha = \frac{32}{27} \nu_\alpha$$

Problem 26.1

For hydrogen molecule, the binding energy is

$$E_{\infty} - E_{\min} = -27.2\text{eV} - (-31.7\text{eV}) = 4.5\text{eV}$$

the binding energy of H_2^+ is 2.7eV



$$4.5 = x + 2.7 - 13.6$$

$$x = 4.5 - 2.7 + 13.6 = 1.8 + 13.6 = 15.4$$

Problem 26.925% HCl in 1st excited state.

$$\begin{aligned} 25\% &= \frac{e^{-\beta h \nu \left(1 + \frac{1}{2}\right)}}{\sum_{n=0}^{\infty} e^{-\beta h \nu \left(n + \frac{1}{2}\right)}} \\ &= \frac{e^{-\beta h \nu}}{\sum_{n=0}^{\infty} \left(e^{-\beta h \nu}\right)^n} = e^{-\beta h \nu} \left(1 - e^{-\beta h \nu}\right) \end{aligned}$$

$$e^{-\beta h \nu} = 0.5$$

$$\beta h \nu = -\ln 0.5 = 0.69$$

$$k_B T = 1.4 h \nu.$$

$$T = \frac{1.4 \times 0.22 \text{ eV}}{k_B 300 / 300}$$

$$= 1.4 \times 0.22 \times 40 \times 300$$

$$= 3.7 \times 10^3 \text{ K}$$

Problem 26.11 (a) The reduced mass is

$$\mu = \frac{35 \times 39}{35 + 39} \text{u} = 18 \text{u}$$

(b) Transition of rotational-vibrational spectrum (for emission) $r_0 = 0.267$

$$\begin{aligned} \Delta E &= E_{N,L} - E_{N-1,L\pm 1} \\ &= \begin{cases} h\nu + \frac{\hbar^2}{mR^2}(-L-1) \\ h\nu + \frac{\hbar^2}{mR^2}L \end{cases} \end{aligned}$$

$$\begin{aligned}\frac{\hbar^2}{mR^2} &= \frac{(hc)^2}{4\pi^2 mc^2 R^2} \\ &= \frac{(1240 \text{ eV} \cdot \text{nm})^2}{4\pi^2 (18 \text{ u} \times 931.5 \text{ MeV/u})(0.267 \text{ nm})^2} \\ &= 3.26 \times 10^{-5} \text{ eV}\end{aligned}$$

Problem 27.2 ($n = 8.60 \times 10^{28} / \text{m}^3$ not given!)

$$\begin{aligned} E_{\text{F}} &= \frac{\hbar^2}{2m} \left(3\pi^2 \times n \right)^{2/3} \\ &= \frac{\left(1240 \times 10^{-9} \right)^2}{8 \times 0.511 \times 10^6} \left(\frac{3 \times 86.0}{\pi} \times 10^{27} \right)^{2/3} \\ &= 7.13 \text{eV} \end{aligned}$$

Problem 27.4 $E_F = 3.00\text{eV}$, Find the probability of energy between 5.00eV to 5.10eV for $T = 295\text{K}$, 2500K

$$P = \frac{gV}{4\pi^2 N} \left(\frac{2m}{\hbar} \right)^{3/2} \int_{5.00\text{eV}}^{5.10\text{eV}} \frac{\varepsilon^{1/2}}{e^{\beta(\varepsilon-\mu)} + 1} d\varepsilon$$
$$= \frac{3}{2E_F^{3/2}} \int_{5.00\text{eV}}^{5.10\text{eV}} \frac{\varepsilon^{1/2}}{e^{\beta(\varepsilon-\mu)} + 1} d\varepsilon$$

$$\varepsilon \sim 5.00\text{eV} - 5.10\text{eV}; \quad \varepsilon - \mu \sim 2.00 - 2.10 \sim 2.05$$

For 295K

$$\begin{aligned}\beta \times 2.05 &= \frac{2.05}{k_B 295} \\ &= \frac{2.05}{k_B 300 \times \frac{295}{300}} \\ &= \frac{2.05 \times 40 \times 300}{295} = 83.4\end{aligned}$$

$$e^{83.4} = 1.64 \times 10^{36} \gg 1$$

$$P \approx \frac{3}{2 \times 1.64} \times 10^{-36} \times \int_{1.67}^{1.70} x^{1/2} dx = 3.56 \times 10^{-38}$$

For 2500K

$$\begin{aligned}\beta \times 2.05 &= \frac{2.05}{k_B 2500} \\ &= \frac{2.05}{k_B 300 \times \frac{2500}{300}} \\ &= \frac{2.05 \times 40 \times 300}{2500} = 9.84\end{aligned}$$

$$e^{9.84} = 1.88 \times 10^4 \gg 1 \quad P = 3.11 \times 10^{-6}$$

Better results are 4.37×10^{-36} and 6.00×10^{-6} .

Sommerfeld expansion may be appealed

$$E = \frac{3}{5} N E_F \left[1 + \frac{5\pi^2}{12} \left(\frac{k_B T}{E_F} \right)^2 + \dots \right]$$

$$\mu = E_F \left[1 - \frac{\pi^2}{12} \left(\frac{k_B T}{E_F} \right)^2 + \dots \right]$$